Project Report

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# Title: DigiConnect

Word count: 480

DATE: 13 March 2026

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# 1. What it does

*The theme I have chosen for my project is Accessibility & Inclusive Design.* [1]

In Ireland, digital exclusion affects more than 1 in 4 Irish adults who fall below basic digital skills. A published report by Accenture found that 42% of Irish people describe themselves as 'average' or below for digital skills. [2] And it is estimated that nearly 30% of adults aged 60-74 have never used the internet.

This is a growing problem, and it means that many find themselves being locked out of essential services such as medical appointments, social welfare, online banking etc. And many people feel too embarrassed to seek help due to the stigma surrounding the issue.

**DigiConnect** is a web app developed to aid in the coordination at a Docklands community hub to register residents who need digital support. The system assesses users' individual skills and matches them to appropriate workshops and one-to-one sessions. The system tracks users progress from beginner, developing and independent. And tracks anyone who has dropped out.

# 2. Inheritance & Poloymorphism

Resident is an abstract base class with three subclasses:

- Beginner: adds device type (e.g. smartphone, laptop, tablet, PC)
- Developing: adds targetSkill (e.g. video calls, online banking)
- Independent: add completedProgram (e.g. DigiConnect 10-week course)

# 3. Three ADT's

The three ADT's include:

- ResidentList – an ArrayList for storing all the registered residents.
- AssesmentQueue – a Queue backed by a SinglyLinkedList. Residents wait here in FIFO order until assessed.
- CompletedSessionStack – a Stack that stores residents who have finished their programme. Most recent completioed appear list (LIFO)

# 4. Singly Linked List

The Singly Linked List backs the AssesmentQueue. Each node holds one resident and a pointer to the next node. The list contains addLast(), removeFirst(), peekFirst() and isEmpty().

# 5. CRUD Operations on 2 ADT's

### 5.1.1 ADT 1: ResidentList (ArrayList)

- Create – register a new resident, add them to the queue
- Read – view all residents in the dashboard, peek at the next in queue
- Update – edit a resident's details or skill tier after assessment
- Delete – remove a duplicate or incorrect registration

### 5.1.2 ADT 2: AssesmentQueue (SinglyLinkedList)

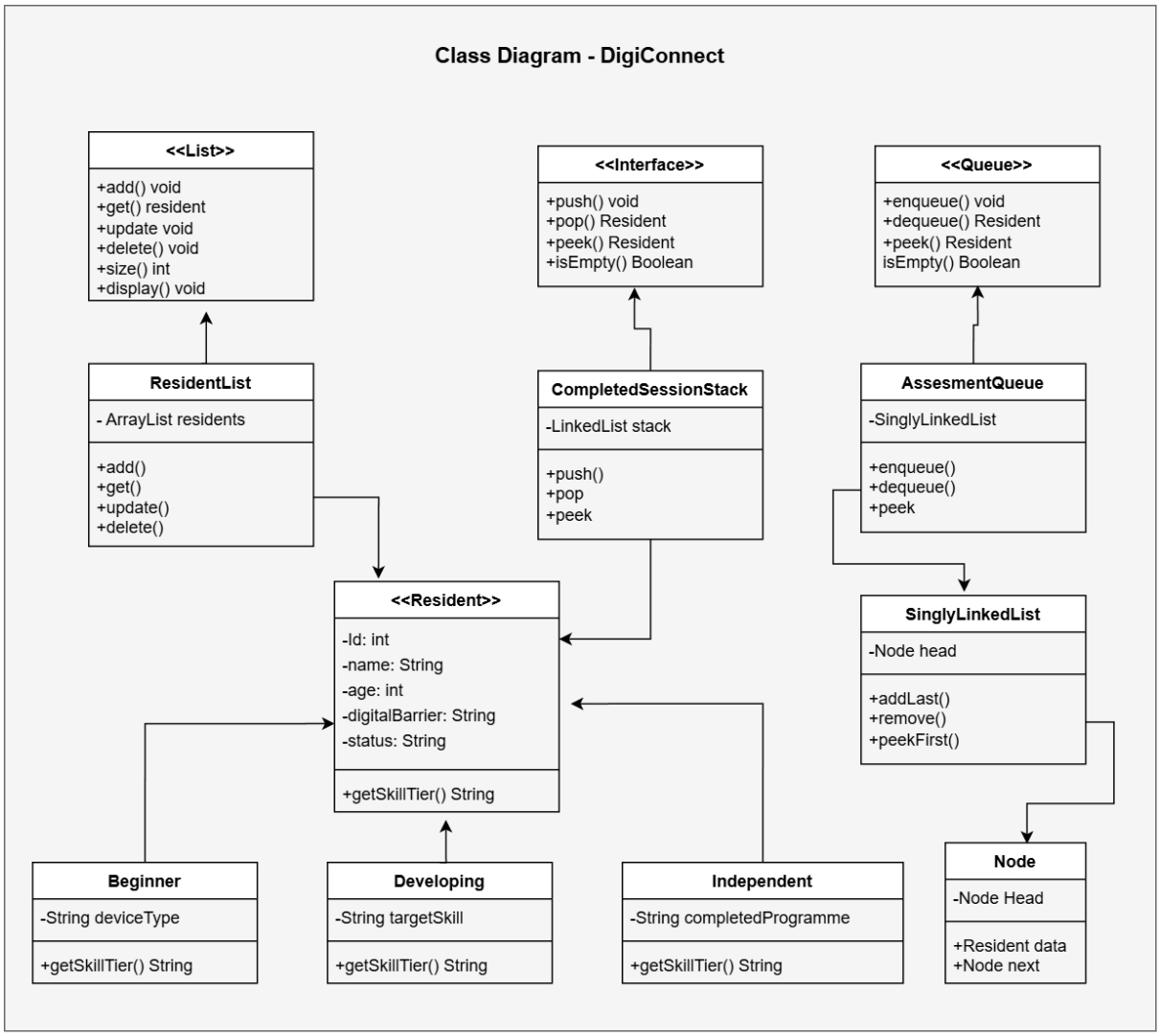
- Create – enqueue a resident when they register, add them to the back of the queue.
- Read – peek at the next resident in the queue without removing them
- Update – re-enqueue a resident if their assessment needs to be rescheduled

- Delete – dequeue a resident from the front when their assessment begins

# 6. Graphical User Interface (GUI)

- The GUI will be created in NetBeans. It will contain the following:
- Main Dashboard – summary counts of waiting, active and completed residents
- Register Resident Form – enter name, age, digital difficulty, and skill tier
- Assessment Queue Panel – view and process the next resident in the queue
- Sesson Tracker Panel – log session attendance and update residents' status
- Progress Dashboard – view completed residents and flag dropouts for follow-up

# Class Diagram




## References:

[1] Smartdoc, ‘Smart Docklands Call for Pilots 2025: Deadline Extended’, smartdocklands. Accessed: Mar. 13, 2026. [Online]. Available: https://smartdocklands.ie/smart-docklands-call-for-pilots-2025-deadline-extended/

[2] julia, ‘Quarter of Irish people feel excluded from digital society’, TechCentral.ie. Accessed: Mar. 13, 2026. [Online]. Available: https://www.techcentral.ie/quarter-of-irish-people-feel-excluded-from-digital-society/